INCH POUND
MIL-DTL-83413/7E
w/AMENDMENT 2
18 April 2014
SUPERSEDING
MIL-DTL-83413/7E
w/AMENDMENT 1
17 June 2013

DETAIL SPECIFICATION SHEET

CONNECTORS AND ASSEMBLIES, ELECTRICAL, AIRCRAFT GROUNDING: GROUNDING CLAMP CONNECTOR FOR TYPES I AND III GROUNDING ASSEMBLIES, ALUMINUM CLIP, ELECTRICAL

This specification is approved for used by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83413.

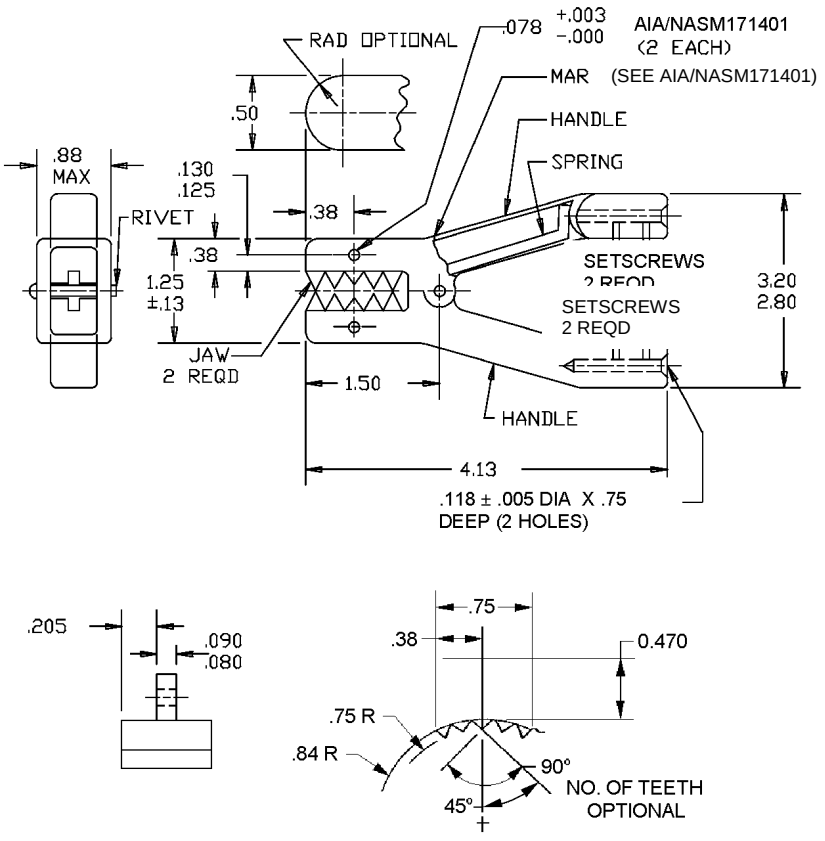


FIGURE 1. Grounding clamp, for types I and III grounding assemblies.

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Inches	mm	Inches	mm	Inches	mm
.003	0.08	.113	2.87	.75	19.1
.004	0.10	.125	3.18	.84	21.3
.010	0.25	.130	3.30	.88	22.4
.031	0.79	.130	3.30	1.25	31.4
.062	1.57	.203	5.16	1.50	38.1
.078	1.98	.205	6.35	2.80	76.1
.080	2.03	.380	9.70	3.20	81.3
.090	2.29	.500	12.70	4.13	104.9

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Surface toughness: If cast, smooth cast finish 125 or 250 all over, except machine surface 63.
4. Remove all burrs and sharp edges .010 (0.25 mm) to .020 (0.51 mm) radius.
5. Unless otherwise specified, tolerance is ±.03 (0.8 mm) on decimals and ± 1° on angles.

FIGURE 1. Grounding clamp, for types I and III grounding assemblies - Continued.

REQUIREMENTS:

Material-handle: Aluminum alloy in accordance with ASTM-A385, ASTM-B26/B26M, ASTM-B108/B108M and SAE-J452, SAE-J453, SAE-AMS-4290, SAE-AMS-4291, Alloy 360, A380 or 384.

Material-spring: .125 diameter music wire in accordance with ASTM-A228/A228M, or equal. Heat treatment shall be as required to meet requirements. The spring shall be stress relieved, zinc/clear plated in accordance with ASTM-B633, or Nickel plated in accordance with ASTM-B733.

Material set screw: AIA/NASM51021, passivated in accordance with SAE-AMS 2700.

Clamp connector resistance: Shall not exceed 500 milliohms.

Salt spray: 96 hours in accordance with test procedure EIA/ECA-364-26 test condition A.

Part or Identifying Number (PIN): M83413/7-1.

Alternate jaw material: MPIF FX - 1008-110HT in accordance with Metal Powder Industries Federation, 105 College Road East, Princeton, New Jersey 08540.

QUALIFICATION:

Qualification is not required. In addition to the stated requirements above, first article testing shall consist of dimensional verification and resistance testing.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-83413, this document references the following:

ASTM-A228/A228M	ASTM-B26/B26M
ASTM-B633	ASTM-B108/B108M
ASTM-B733	SAE-J452
EIA/ECA-364-26	SAE-J453
SAE-AMS 2700	SAE-AMS-4290
ASTM-A385	SAE-AMS-4291

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5999-2014-002)

Review activities:
Army - AR, CR, CR4, MI
Navy - MC
Air Force - 02, 13, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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